Sahaj Bhadja

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Summary

Junior software engineer with 2+ years building **Python/Java services**, React interfaces, relational data pipelines, automated tests, and cloud deployments; experienced with distributed failure handling, Azure, Docker, Kubernetes, and eager to develop in C#/.NET.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python FastAPI services and relational schemas** for 500+ users, implementing row-level isolation, indexed queries, and optimized access patterns that reduced analytical dashboard latency from 1.2 seconds to 350 milliseconds.
- Developed **fault-tolerant data pipelines** processing 5,000+ monthly jobs with incremental loads, validation, reconciliation, explicit job states, failure isolation, and CI/CD deployments, achieving 99%+ reliability across production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented **React interfaces and Python Flask APIs** for upload, processing, and history workflows, supporting 150+ users and 10,000+ daily requests while maintaining stable performance under production load.
- Engineered **secure ingestion and concurrent processing** for PDF, DOCX, and ZIP inputs with validation, persistent results, fallback handling, and fault-tolerant execution, processing 1,000+ files per batch reliably.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python data and evaluation pipelines** processing 100K+ observational trajectory records, integrating spatial-temporal features and reproducible simulation tests that improved long-horizon model performance 18% across benchmark scenarios.
- Investigated **model behavior and failure patterns** across three architectures, documenting benchmark metrics and diagnostic findings that guided engineering improvements and increased closed-loop planning stability by 15% in simulation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Troubleshot and optimized **API and ETL integrations** using OOP, AWS Glue, S3, SQL, caching, JSON, and Parquet transformations, improving response times 30%, pipeline efficiency 68%, and storage costs 15%.
- Automated **builds, deployments, and 20+ unit/integration suites** with GitHub, Jenkins, PyTest, and Mockito, generating coverage and failure reports while supporting reliable releases across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **React interfaces and Node.js APIs** across eight microservices backed by Azure SQL, synchronizing 2M+ records through WebSockets while supporting authentication, resource, billing, permissions, and notification workflows.
- Added **unit, integration, and end-to-end tests**, Docker packaging, and Jenkins CI/CD with staged approvals, while using Sentry, Azure Monitor, and Kibana to diagnose errors, metrics, and production logs.

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, Bash; **Learning**: C#, .NET, ASP.NET Core, Blazor
Full Stack: FastAPI, Flask, Node.js, Express.js, React.js, Next.js, REST APIs, WebSockets
Data: PostgreSQL, MySQL, SQL Server, Azure SQL, relational modeling, schema design, query tuning, ETL
Testing & Observability: PyTest, Mockito, Jest, Selenium, Sentry, Azure Monitor, Kibana, validation, error handling
Platform: Azure Blob Storage, Docker, Kubernetes, Git, GitHub Actions, Jenkins, CI/CD, distributed systems

Projects

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed a **containerized Python data-processing API** on AWS ECS/Fargate with CodePipeline, automated validation checks, and CI/CD releases, achieving 87% accuracy and production-grade operational reliability at scale.

HMIS Live | *TypeScript, MERN, REST APIs, Sentry*
- Built a **full-stack operational platform** with TypeScript dashboards, REST integrations, authentication, appointment workflows, Twilio notifications, and Sentry monitoring, reducing administrative workload 30% while improving production issue visibility.

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes*
- Built a **real-time data platform** supporting 50,000+ users and 1M+ transactions, combining responsive Next.js interfaces, REST integrations, WebSockets, MongoDB access, and Kubernetes-based CI/CD automation for scalable operations.